

Thermodynamics

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

Q1 JEE Main 2020 - 4 September (Morning)

For one mole of an ideal gas, which of these statements must be true?

- (a) U and H each depends only on temperature
- (b) Compressibility factor z is not equal to 1
- (c) $C_{P,m} - C_{V,m} = R$
- (d) $dU = C_V dT$ for any process

- (A) (a) and (c)
- (B) (a), (c) and (d)
- (C) (b), (c) and (d)
- (D) (c) and (d)

Q2 JEE Main 2020 - 4 September (Evening)

Five moles of an ideal gas at 1 bar and 298K is expanded into vacuum to double the volume. The work done is

- (A) Zero
- (B) $-RT \ln V_2/V_1$
- (C) $C_V (T_2 - T_1)$
- (D) $-RT(V_2 - V_1)$

Q3 JEE Main 2020 - 5 September (Evening)

For a dimerization reaction,



at 298K, $\Delta U^\circ = -20 \text{ kJ mol}^{-1}$, $\Delta S^\circ = -30 \text{ JK}^{-1} \text{ mol}^{-1}$, then the ΔG° will be J.

Q4 JEE Main 2020 - 6 September (Morning)

The variation of equilibrium constant with temperature is given below:

Temperature	Equilibrium Constant
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$T_1 = 25^\circ\text{C}$	$K_1 = 10$
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$T_2 = 100^\circ\text{C}$	$K_2 = 100$
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Thermodynamics

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Questions with Answer Keys

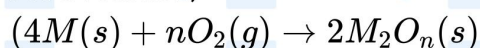
Chemistry

The values of ΔH° , ΔG° at T_1 and ΔG° at T_2 (in kJ mol^{-1}) respectively, are close to [Use $R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$]

- (A) 28.4, -5.71 and -14.29
- (B) 0.64, -7.14 and -5.71
- (C) 28.4, -7.14 and -5.71
- (D) 0.64, -5.71 and -14.29

Q5 JEE Main 2020 - 6 September (Evening)

For a reaction,

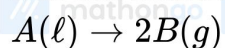


the free energy change is plotted as a function of temperature. The temperature below which the oxide is stable could be inferred from the plot as the point at which

- (A) the free energy change shows a change from negative to positive value
- (B) the slope changes from positive to negative
- (C) the slope changes from negative to positive
- (D) the slope changes from positive to zero

Q6 JEE Main 2020 - 7 January (Morning)

For the reaction :

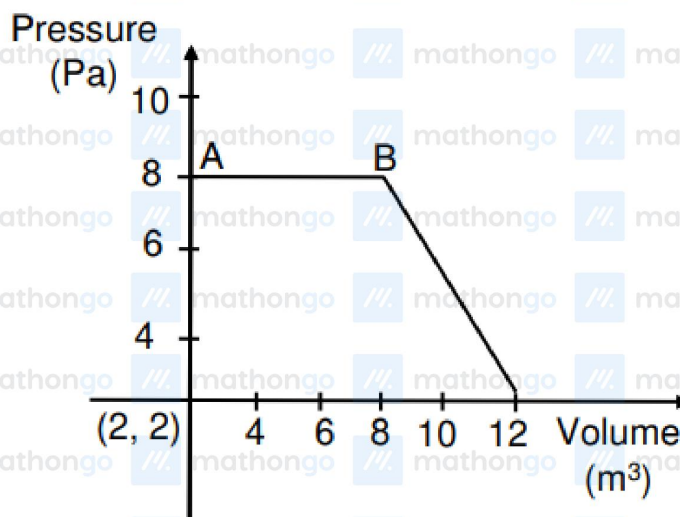


$\Delta U = 2.1 \text{ kcal}$, $\Delta S = 20 \text{ cal/K}$, at $T = 300 \text{ K}$.

Hence ΔG (in kcal) is $-X$. Find the value of X

Q7 JEE Main 2020 - 8 January (Morning)

The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC



shown in the figure is

Q8 JEE Main 2020 - 8 January (Evening)

At constant volume, 4 mol an ideal gas when heated from 300 K to 500 K changes its internal energy by 5000 J . The molar heat capacity at constant volume is. . .

Multiply your answer with 100

Q9 JEE Main 2020 - 9 January (Evening)

The true statement amongst the following is:

- (A) Both S and ΔS are not functions of temperature.
- (B) S is not a function of temperature but ΔS is a function of temperature.
- (C) Both ΔS and S are functions of temperature.
- (D) S is a function of temperature but ΔS is not a function of temperature.

Answer Key

Q1 (B)

Q2 (A)

Q3 (-13537.57)

Q4 (A)

Q5 (A)

Q6 (-2.7)

Q7 (48)

Q8 (625)

Q9 (C)

Thermodynamics

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Q1

For 1 mole of ideal gas.

- Both internal energy (U) and Enthalpy (H) depends on temperature

Compressibility factor $Z = 1$

$$\bullet C_{P,m} - C_{v,m} = R$$

$$\text{and } dU = C_v dT$$

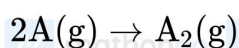
Q2

$$P_{\text{ext}} = 0$$

$$w = -P_{\text{ext}} \Delta V$$

$$= 0$$

Q3



$$\Delta U^\circ = -20 \text{ kJ/mol and } \Delta S^\circ = -30 \text{ J/Kmol}$$

$$\Delta H^\circ = \Delta U^\circ + \Delta n_g RT$$

$$= -20000 - 1 \times 8.314 \times 298$$

$$= -22477.57$$

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$= -22477.57 + 298 \times 30$$

$$= -13537.57$$

Q4

$$\Delta G^\circ = -RT \ln K$$

$$T_1 = 25^\circ \text{C} \quad K_1 = 10$$

$$\Delta G^\circ \text{ at } T_1 = -8.314 \times 298 \times 2.303 \times \log 10$$

$$= -5.71 \text{ kJ/mol}$$

$$\Delta G^\circ \text{ at } T_2 = -8.314 \times 373 \times 2.303 \times \log(100)$$

$$= -14.29 \text{ kJ/mol}$$

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$-5.71 = \Delta H^\circ - 298 (\Delta S^\circ)$$

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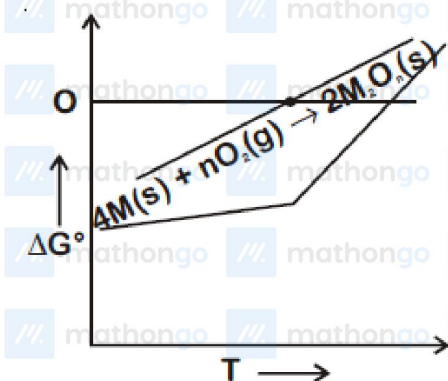
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$$-14.29 = \Delta H^\circ - 373 (\Delta S^\circ)$$

$$\Delta H^\circ = 28.4 \text{ kJ/mol}$$

Q5

From Ellingham Diagram



From graph it is evident, that the temperature below which the oxide is stable, is the point at which free energy change shows a change from negative to positive.

Q6

$$\Delta H = \Delta U + \Delta n_g RT$$

$$= 2.1 \times 10^3 + 2(2)(300)$$

$$= 2100 + 1200$$

$$= 3300 \text{ cal}$$

$$\Delta G = \Delta H - T\Delta S$$

$$= 3300 - (300)(20)$$

$$= 3300 - 6000$$

$$= -2700 \text{ cal} = -2.7 \text{ kcal}$$

Q7

$$|w| = \frac{1}{2}(6 + 10) \times 6 = 48 \text{ J}$$

Q8

Thermodynamics

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

$$\Delta U = nC_v \Delta T$$

$$5000 = 4 \times C_v (500 - 300)$$

$$C_v = 6.25 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$\text{Final ans} = 6.25 \times 100 = 625 \text{ JK}^{-1} \text{ mol}^{-1}$$

Q9

$$\Delta S = \int \frac{dq}{T}$$

$$S_T = \int_0^T \frac{nCd T}{T}$$